Python-Based SSH Log Analyzer

Automated Detection of Failed Authentication Attempts

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Tools used:

- Kali Linux
- Virtual Box
- OpenSSH
- Python 3
- journalctl
- grep

1. Introduction

This lab demonstrates the development of a Python-based log analysis tool designed to detect failed SSH authentication attempts. Instead of manually reviewing logs using command-line tools, a custom script was created to automate parsing and summarizing security events.

The objective was to simulate real-world Security Operations Center (SOC) analyst responsibilities by detecting suspicious login patterns programmatically.

2. Lab Environment

- Operating System: Kali Linux (Virtual Machine)
- Service Analyzed: OpenSSH (sshd)
- Log Source: Exported SSH journal logs
- Programming Language: Python 3
- Tools Used:
    - journalctl
    - grep
    - Python (re, collections modules)

3. Objective

The primary objectives of this lab were:

- Export SSH authentication logs
- Develop a Python script to parse log data
- Identify failed login attempts
- Extract usernames and IP addresses
- Quantify suspicious activity
- Generate a structured analysis report

4. Methodology

Step 1: Generate Failed Login Attempts

Multiple failed SSH login attempts were simulated using a non-existent user account. This generated authentication failure entries within the system logs.

Step 2: Export SSH Logs

Failed authentication logs were exported using:

```
sudo journalctl -u ssh | grep "Failed" > failed_attempts.txt
```

This created a log file for analysis.

Step 3: Develop Python Log Analyzer

A Python script was developed to:

- Read the log file
- Identify lines containing “Failed password”
- Extract IP addresses using regular expressions
- Extract targeted usernames
- Count total failed attempts
- Summarize attempts per IP address

The script uses:

- re module for pattern matching
- collections. Counter for frequency analysis

5. Findings

The Python log analyzer detected:

- Total Failed Attempts: 13
- Unique Source IP Address: 127.0.0.1

- Targeted Username: fakeuser
- Attempts Per IP: 13 attempts from 127.0.0.1

The repeated failed authentication attempts simulate brute-force login behavior. The script successfully automated detection and summary of suspicious authentication events.

6. Security Implications

Repeated authentication failures may indicate:

- Brute-force attacks
- Credential guessing attempts
- Automated attack scripts

Automated log parsing enables faster identification of such activity and reduces manual investigation time.

7. Recommendations

To mitigate SSH brute-force risks:

- Enable Fail2Ban
- Implement SSH key-based authentication
- Disable password authentication
- Restrict SSH access by IP
- Continuously monitor authentication logs

8. Conclusion

This lab successfully demonstrated the development of an automated log analysis tool using Python. The project highlights the importance of automation in cybersecurity operations and reflects practical SOC analyst responsibilities.

The script efficiently parsed authentication logs, identified suspicious patterns, and generated actionable summaries.